AI BASED QUESTION BANK

PREDICTION SYSTEM

Based on NCERT Curriculum

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Program: MCA 3rd Semester

Introduction

Education is evolving rapidly with the integration of Artificial Intelligence. Traditional quiz and question bank systems are static and inefficient. This project presents an AI-powered web application that automatically generates quiz questions from educational PDFs — specifically aligned with the NCERT curriculum.

Problem

Manual question creation is time-consuming and lacks personalization for each student.

Solution

AI reads PDF content and auto-generates MCQ-based quizzes using Gemini AI API.

Goal

Build a smart, scalable quiz system for Class 6–10 NCERT subjects with instant results.

Project Objectives

01 Develop an AI-powered web application using HTML, CSS, JavaScript, and Node.js.

02 Integrate Gemini AI API to automatically generate MCQ questions from NCERT PDF content.

03 Enable class-wise student registration, login, and quiz attempt tracking.

04 Provide Admin Panel to manage students, subjects, and view performance analytics.

05 Show instant, detailed quiz results with correct answers and score summary.

06 Support 11 questions per quiz with real-time progress tracking and timer.

Technology Stack

Frontend

HTML5 · CSS3 · JavaScript (ES6+)

Backend

Node.js · Express.js · REST API

AI Engine

Google Gemini AI API (Question Generation)

PDF Processing

pdf-parse · multer (File Upload)

Database

JSON File Storage / Local DB

Auth & Sessions

bcrypt · express-session · cookie-parser

System Architecture & Workflow

Admin uploads NCERT PDF

PDF parsed by pdf-parse

Gemini AI generates MCQ questions

Questions saved to Question Bank

Student takes the quiz

Instant result & analysis

Key System Components

- Home Page with Student Register / Login / Admin Panel
- PDF Reader module for extracting text from NCERT books
- AI Question Generation engine via Gemini API
- Detailed Results module with score breakdown

Features

AI Question Generation

Gemini AI reads NCERT PDF and auto-creates 11 MCQ questions per quiz session.

Student Auth System

Secure student registration, login with class-wise access (Class 6–10).

Detailed Results

Instant result page with score, correct answers highlighted, and performance summary.

Admin Panel

Admins can manage students, upload PDFs, monitor quiz attempts, and view analytics.

Progress Tracking

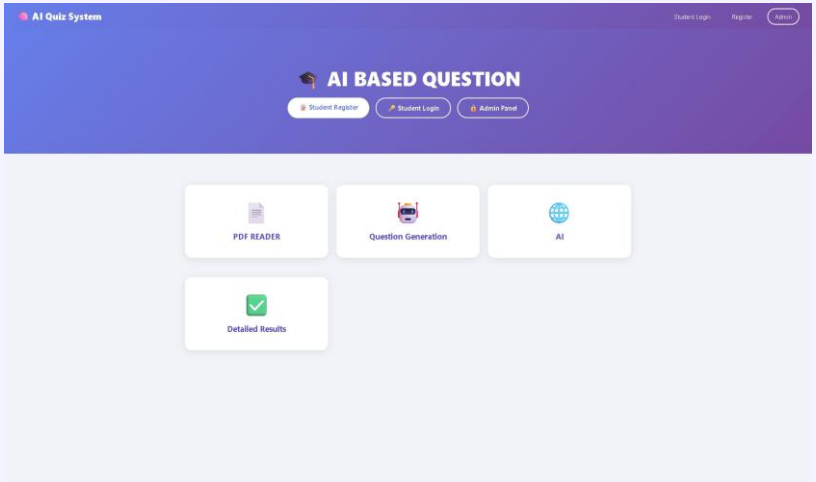
Real-time quiz progress bar showing questions answered out of total (e.g. 0/10).

Responsive UI

Clean, modern interface that works across desktop and mobile browsers.

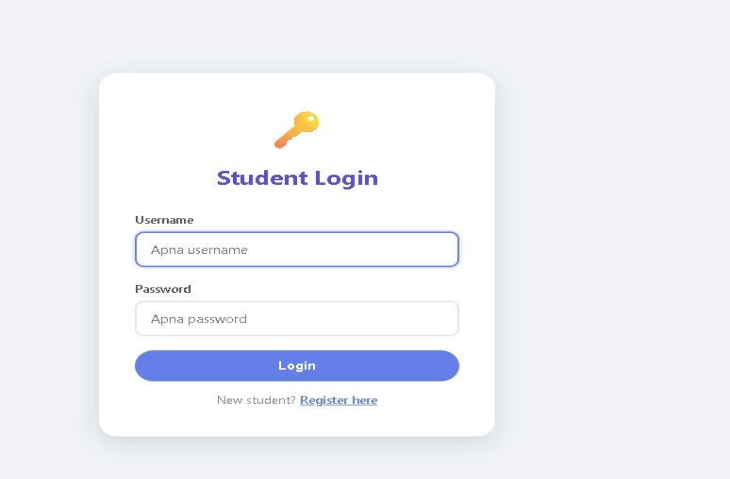
User Access points

Home Page



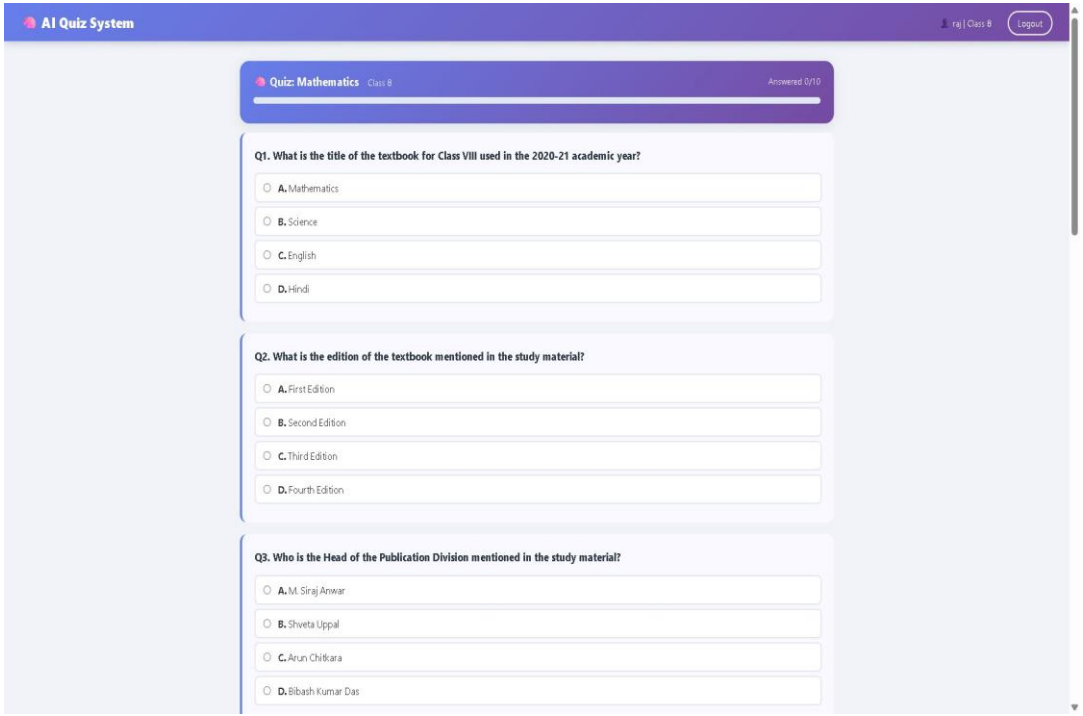
Home page with Student Register, Login, and Admin Panel access with 4 core feature modules.

Student Login



Secure student login with Username & Password authentication. New students can register directly.

Quiz Interface



- Progress bar shows answered questions (e.g., 0/10)
- Class-specific quiz (e.g., Mathematics – Class 6th to 12th)
- 11 MCQ questions auto-generated by AI from PDF
- 4 options (A, B, C, D) per question with radio buttons
- Student name and class shown in top-right corner
- Logout available anytime during the quiz session

Future Scope & Conclusion

Future Enhancements

- Multi-language question generation support
- Mobile app version (Android / iOS)
- AI-based difficulty level prediction
- Parent dashboard for student monitoring
- Cloud database integration (MongoDB / Firebase)
- Gamification — badges and leaderboards

Conclusion

This project successfully demonstrates how Artificial Intelligence can transform the traditional education system.

Using Node.js, HTML/CSS, and the Gemini AI API, the system provides an intelligent, scalable, and user-friendly quiz platform fully aligned with the NCERT curriculum.

It bridges the gap between manual question creation and automated, personalized learning — making quality education more accessible.

Thank You!

for your attention

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